MIL-S-46897A(MI)
29 April 1983
SUPERSEDING
MIL-P-46897(MI)
14 December 1973

MILITARY SPECIFICATION

SEALING COMPOUND,

POLYURETHANE FOAM

This specification is approved for use by US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers one type of polyurethane foam sealing compound used as a nozzle seal in rocket motors.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

	PPP-B-601	Boxes, Wood, Cleated-Plywood
*	PPP-B-621	Box, Wood, Nailed and Lock Corner
	PPP-B-636	Boxes, Shipping, Fiberboard
	PPP-C-96	Can, Metal, 28 Gage and Lighter

*

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-RSDS, Redstone Arsenal, AL 35898 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8030

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STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
* MIL-STD-1188	Commercial Packaging of Supplies and Equipment

* (Copies of specifications, standards, handbooks, drawings, and publications, required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

American Society for Testing and Materials

ASTM D 1622	Test for Apparent Density of Rigid Cellular plastics
ASTM D 1638	Testing Urethane Foam Isocyanate Raw Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

* (Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. The material shall be a rigid, closed-cell polyurethane foam, and shall be a self-curing, two-component system of urethane resin and isocyanate curing agent.

3.2 Chemical and physical properties. The chemical and physical properties shall be as specified in Table I.

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* Table I. Chemical and physical properties

Property	Minimum	Maximum	Test Method
Resin component			
Specific gravity	1.22	1.30	4.3.2
Hydroxyl number	455	515	4.3.3
Isocyanate component			
Specific gravity	1.12	1.20	4.3.2
Isocyanate content (percent)	32	38	4.3.4
Foam			
Bulk density, lb/ft^3	4.5	7.5	4.3.5

* 3.3 Workmanship. The sealing compound components shall be uniform in quality and free from impurities and other defects that could adversely affect their use.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. All inspections specified herein are classified as quality conformance inspections.

4.2.1 Inspection lot. An inspection lot shall consist of material from one batch or run subjected to the same processing operations and conditions by the same manufacturer, and presented for inspection at one time.

* 4.2.2 Sampling. Sampling for visual examination shall be 100 percent of the lot. Sampling for all other examinations and tests shall be a representative sample selected at random. Failure of the sample to meet any requirement of this specification shall cause rejection of the lot.

4.2.3 Procedure. Samples taken in accordance with 4.2.2 shall be subjected to the examinations and tests as listed in Table II.

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* Table II. Examinations and Tests

Examination	Test Paragraph	Requirement Paragraph
Visual	4.3.1	3.1, 3.3, 5
Specific gravity	4.3.2	3.2
Hydroxyl number	4.3.3	3.2
Isocyanate content	4.3.4	3.2
Bulk density	4.3.5	3.2

4.3 Inspection methods. The inspection methods shall be as follows:

4.3.1 Visual examination. Visual examination of the samples specified in 4.2.2 shall be conducted to determine compliance with 3.1, 3.3, and 5.

4.3.2 Specific gravity. The specific gravity of the component materials shall be determined in accordance with ASTM D 1638.

4.3.3 Hydroxyl number. The hydroxyl number of the resin component shall be determined in accordance with ASTM D 1638.

4.3.4 Isocyanate content. The isocyanate content of the curing agent shall be determined in accordance with ASTM D 1638.

4.3.5 Bulk density. The bulk density of the cured foam shall be determined in accordance with ASTM D 1622.

* 5. PACKAGING

5.1 Preservation. Preservation, packaging, packing, unitization and marking shall be Level A, B or Industrial (see 6.2).

5.1.1 Level A. Polyurethane shall be packaged in containers conforming to PPP-C-96 or an Industrial type container suitable for polyurethane foam.

5.1.2 Level B. Same as Level A.

5.1.3 Industrial. Preservation and packaging shall be in accordance with MIL-STD-1188.

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* 5.2 PACKING

5.2.1 Level A. Packaged polyurethane shall be packed in snug-fitting containers conforming to PPP-B-601, overseas type or PPP-B-621 Class 2 as applicable.

5.2.2 Level B. Packaged polyurethane shall be packed in fiberboard containers conforming to PPP-B-636, Class weather resistant.

5.2.3 Industrial. Polyurethane shall be packed in accordance with the requirements of MIL-STD-1188.

* 5.3 MARKING

5.3.1 Level A and B. Marking shall be in accordance with MIL-STD-129. Additional marking shall include, but not be limited to the following:

a. Manufacturer's name and location.

b. Material trade name.

c. Net weight.

d. Lot number, batch number, and date of manufacture.

e. Number and revision letter of this specification.

f. Shelf life or storage limitations.

5.3.2 Industrial. Marking shall be in accordance with MIL-STD-1188. Additional markings shall include but not be limited to the following:

a. Manufacturer's name and location.

b. Material trade name.

c. Net weight.

d. Lot number, batch number, and date of manufacture.

e. Number and revision letter of this specification.

f. Shelf life or storage limitations.

6. NOTES

* 6.1 Intended use. This type of sealing compound is intended for use as a nozzle seal in rocket motors.

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* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and revision letter of this specification.

b. Minimum lot size, if applicable.

c. Place of delivery.

d. Size of container, or foam sheet.

e. Place of inspection.

f. Disposition of test data.

* g. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

* 6.3 Supersession data. This specification includes the requirements of Purchase Description, MIS-15294, dated 18 February 1965.

* The margins of this revision are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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